Lab 10 – Worksheet

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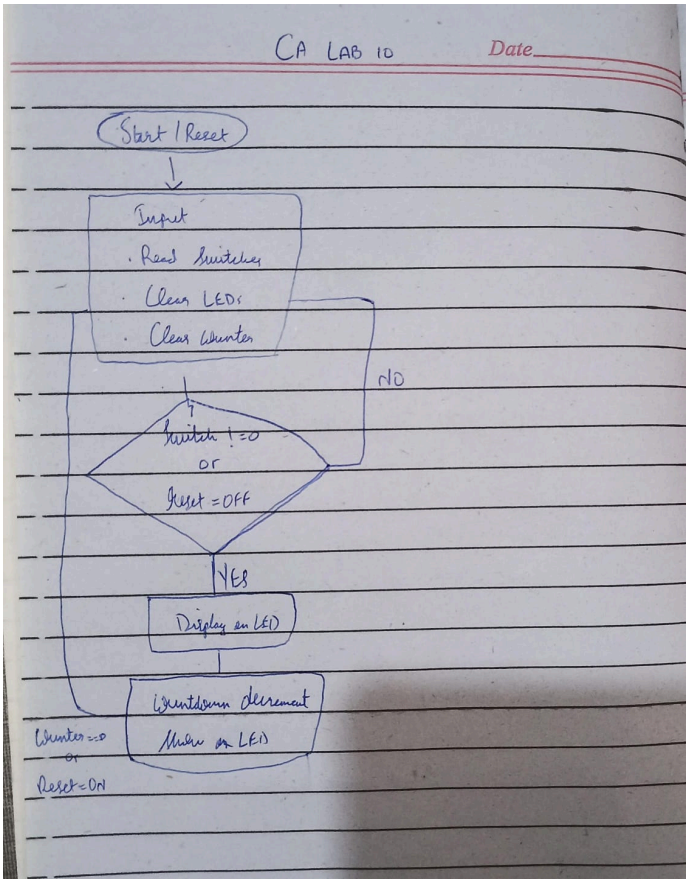
Note: Assumptions and logics should be explained separately in tasks after the task results.

Task 1

FSM of assembly implementation of FSM designed in Lab 05



Task 2

Complete assembly source file implementing the FSM.

```
.text
.globl _start
_start:
li sp, 0x1F0 # init stack pointer
addi s1, zero, 512 # s1 = LED address
addi s2, zero, 768 # s2 = switch address
addi s3, zero, 772 # s3 = reset button address
INPUT_WAITING:
sw zero, 0(s1) # clear LEDs
POLL_SWITCHES:
lw a0, 0(s2) # read switches
beq a0, zero, POLL_SWITCHES # wait until non-zero
jal ra, COUNTDOWN_SUB # a0 = captured switch value
j INPUT_WAITING
# arg: a0 = initial count value
# stack frame: sp+4 = ra, sp+0 = s0
COUNTDOWN_SUB:
addi sp, sp, -8
sw ra, 4(sp)
sw s0, 0(sp)
mv s0, a0 # s0 = counter
COUNTDOWN_LOOP:
beq s0, zero, COUNTDOWN_DONE
lw t0, 0(s3) # check reset
bne t0, zero, RESET_DETECTED
sw s0, 0(s1) # update LEDs
jal ra, DELAY
addi s0, s0, -1
j COUNTDOWN_LOOP
RESET_DETECTED:
sw zero, 0(s1) # clear LEDs
j COUNTDOWN_DONE
COUNTDOWN_DONE:
sw zero, 0(s1) # clear LEDs
lw ra, 4(sp) # restore frame
lw s0, 0(sp)
```

```
addi sp, sp, 8
ret
# busy-wait: 1000 x 500 x ~2 cycles = ~100ms at 10MHz
# leaf function — does not touch stack
DELAY:
addi t1, zero, 1000
DELAY_OUTER:
addi t2, zero, 500
DELAY_INNER:
addi t2, t2, -1
bne t2, zero, DELAY_INNER
addi t1, t1, -1
bne t1, zero, DELAY_OUTER
ret
```

Task 3

Design module code of ***instruction memory module*** *containing memory initialized with assembly instructions:*

```
`timescale 1ns/1ps

module instructionMemory #(
   parameter OPERAND_LENGTH = 31
)(
   input  [OPERAND_LENGTH:0] instAddress,
   output [31:0] instruction
);

   // 32-bit memory array (Word-addressable)
   reg [31:0] rom [0:33];

   initial begin
      // start
      rom[0]  = 32'h1F000113; // li sp, 0x1F0
      rom[1]  = 32'h20000493; // li s1, 0x200 (512)
      rom[2]  = 32'h30000913; // li s2, 0x300 (768)
      rom[3]  = 32'h30400993; // li s3, 0x304 (772)

      // INPUT WAITING
      rom[4]  = 32'h0004A023; // sw zero, 0(s1)

      // POLL SWITCHES
      rom[5]  = 32'h00092503; // lw a0, 0(s2)
      rom[6]  = 32'hFE050EE3; // beq a0, zero, POLL_SWITCHES
      rom[7]  = 32'h00C000EF; // jal ra, COUNTDOWN_SUB
      rom[8]  = 32'hFF1FF06F; // j INPUT_WAITING

      // COUNTDOWN SUB
      rom[9]  = 32'hFF810113; // addi sp, sp, -8
      rom[10] = 32'h00112223; // sw ra, 4(sp)
      rom[11] = 32'h00812023; // sw s0, 0(sp)
      rom[12] = 32'h00050413; // mv s0, a0

      // COUNTDOWN LOOP
      rom[13] = 32'h02040263; // beq s0, zero, COUNTDOWN_DONE
      rom[14] = 32'h0009A283; // lw t0, 0(s3)
      rom[15] = 32'h00029A63; // bne t0, zero, RESET_DETECTED
      rom[16] = 32'h0084A023; // sw s0, 0(s1)
      rom[17] = 32'h024000EF; // jal ra, DELAY
      rom[18] = 32'hFFF40413; // addi s0, s0, -1
      rom[19] = 32'hFE9FF06F; // j COUNTDOWN_LOOP

      // RESET DETECTED
      rom[20] = 32'h0004A023; // sw zero, 0(s1)
```

```
    rom[21] = 32'h0080006F; // j COUNTDOWN_DONE

    // COUNTDOWN DONE
    rom[22] = 32'h0004A023; // sw zero, 0(s1)
    rom[23] = 32'h00412083; // lw ra, 4(sp)
    rom[24] = 32'h00012403; // lw s0, 0(sp)
    rom[25] = 32'h00810113; // addi sp, sp, 8
    rom[26] = 32'h00008067; // ret

    // DELAY
    rom[27] = 32'h3E800313; // li t1, 1000

    // DELAY OUTER
    rom[28] = 32'h1F400393; // li t2, 500

    // DELAY INNER
    rom[29] = 32'hFFF38393; // addi t2, t2, -1
    rom[30] = 32'hFE039CE3; // bne t2, zero, DELAY_INNER
    rom[31] = 32'hFFF30313; // addi t1, t1, -1
    rom[32] = 32'hFE0318E3; // bne t1, zero, DELAY_OUTER
    rom[33] = 32'h00008067; // ret
  end

  // The PC address is byte-aligned (0, 4, 8), so we shift right by 2 to get the array
index
  assign instruction = rom[instAddress[7:2]];

Endmodule
```

Machine code:

```
1F000113    li sp, 0x1F0                # Initialize stack pointer
20000493    li s1, 512                  # s1 = LED base address (0x200)
30000913    li s2, 768                  # s2 = Switch base address (0x300)
30400993    li s3, 772                  # s3 = Reset button address (0x304)

WAIT_LOOP:
0004A023    sw zero, 0(s1)              # Clear LEDs

POLL_SWITCHES:
00092503    lw a0, 0(s2)                # Read switches
FE050EE3    beq a0, zero, POLL_SWITCHES     # Wait until non-zero
00C000EF    jal ra, COUNTDOWN_SUB           # Call countdown subroutine
FF1FF06F    j INPUT_WAITING                 # Return to IDLE state
```

```
COUNTDOWN_SUB:
FF810113    addi sp, sp, -8              # Save context on stack
00112223    sw ra, 4(sp)
00812023    sw s0, 0(sp)
00050413    mv s0, a0                    # Initialize counter with switch value

COUNTDOWN_LOOP:
02040263    beq s0, zero, COUNTDOWN_DONE
0009A283    lw t0, 0(s3)                 # Check reset button
00029A63    bne t0, zero, RESET_DETECTED
0084A023    sw s0, 0(s1)                 # Update LED output
024000EF    jal ra, DELAY                # Call delay subroutine
FFF40413    addi s0, s0, -1              # Decrement counter
FE9FF06F    j COUNTDOWN_LOOP             # Loop

RESET_DETECTED:
0004A023    sw zero, 0(s1)               # Clear LEDs immediately
0080006F    j COUNTDOWN_DONE             # Exit countdown

COUNTDOWN_DONE:
0004A023    sw zero, 0(s1)               # Ensure LEDs are clear
00412083    lw ra, 4(sp)                 # Restore context from stack
00012403    lw s0, 0(sp)
00810113    addi sp, sp, 8
00008067    ret                          # Return to main

DELAY:
3E800313    li t1, 1000                  # Outer loop counter

DELAY_OUTER:
1F400393    li t2, 500                   # Inner loop counter

DELAY_INNER:
FFF38393    addi t2, t2, -1              # Decrement inner
FE039CE3    bne t2, zero, DELAY_INNER
FFF30313    addi t1, t1, -1              # Decrement outer
FE0318E3    bne t1, zero, DELAY_OUTER
00008067    ret
```

Lab 10: State-Based Control Flow Using RISC-V Assembly

Points Distribution

Task No.	LR2 Code	LR 5 Results	LR11 Design	AR7
Task 1	-	-	/20	/20
Task 2	/20	/20		
Task 3	/20	-		
Total Points	/80			/20
CLO Mapped	CLO 1			

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code / Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
AR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.